

Yue Chu

PhD, MSPH, MBBS

Department of Sociology, Institute for Population Research
The Ohio State University
Columbus, Ohio, USA
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RESEARCH AREAS

Population health, demography, epidemiology, global health estimates, mortality modeling, demographic and disease surveillance, AI in global health, data systems, health disparities.

EDUCATION AND TRAINING

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| 2025 | PhD in Sociology , The Ohio State University, Columbus, USA
Minor in Statistics and Demography
<i>Dissertation: Leveraging Language Models and Machine Learning in Verbal Autopsy Analysis</i> |
| 2020 | MA in Sociology , The Ohio State University, Columbus, USA
<i>Thesis: SVD-Bayes: A Singular Value Decomposition-Based Approach Under Bayesian Framework for Indirect Estimation of Age-Specific Fertility and Mortality</i> |
| 2014 | MSPH , Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
Concentration in Population, Family and Reproductive Health
<i>Thesis: The influence of family environment factors on children's consumption of healthy foods among 5-9 years old children from low-income families in New York City</i> |
| 2012 | Bachelor of Medicine , Fudan University, Shanghai, China
Specialty in Preventive Medicine
<i>Thesis: Evaluation of tuberculosis patient management strategies: evidence from a randomized controlled trial</i> |

PROFESSIONAL EXPERIENCE

Feb 2026–present	Postdoctoral Fellow , Institute for Population Research, The Ohio State University, Columbus, OH
May 2025–Jan 2026	Postdoctoral Fellow , Health and Environment Modeling Laboratory, The Ohio State University, Columbus, OH
Aug 2023–Sep 2023	Visiting Scholar , Africa Health Research Institute, Somkele, South Africa
May 2022–May 2023	Data Science Intern , Nationwide Mutual Insurance Company / Nationwide Center for Advanced Customer Insights, Columbus, OH
Jan 2019–May 2025	Graduate Research Associate , Institute for Population Research, The Ohio State University, Columbus, OH
Jan 2015–Jul 2018	Research Associate , Department of International Health, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD
Sep 2014–Jan 2015	Research Assistant , Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health
Apr 2013–Sep 2014	Research Assistant , Johns Hopkins Bloomberg School of Public Health
May 2013–Aug 2013	Research Assistant , Technical Division, United Nations Population Fund (UNFPA), New York
Apr 2012–Jun 2012	Research Assistant , Department of Health Statistics and Social Medicine, School of Public Health, Fudan University, Shanghai, China
Feb 2012–Mar 2012	Intern , Division of Vital Statistics, Center for Disease Control and Prevention, Shanghai, China
Mar 2011–Aug 2011	Clinical Intern , Fifth People's Hospital, Shanghai, China

PUBLICATIONS

Papers

25. **Chu, Y.** Leveraging language models and machine learning in verbal autopsy analysis [preprint]. *arXiv*. 2025. <https://doi.org/10.48550/arXiv.2508.19274>
24. **Chu, Y.**, Marston, M., Dube, A., Festo, C., Geubbels, E., Gregson, S., Herbst, K., et al. Temporal changes in cause of death among adolescents and adults in six countries in eastern and southern Africa in 1995-2019: A multi-country surveillance study of verbal autopsy data. *Lancet Global Health*. 2024;12(8):e1278-e1287. [https://doi.org/10.1016/S2214-109X\(24\)00171-2](https://doi.org/10.1016/S2214-109X(24)00171-2)
23. Damerow, S. M., Yeung, D., Martins, J. S., Pathak, I., **Chu, Y.**, Liu, L., Fisker, A. B. Provider-mother interactions are associated with birth outcome misclassifications in household surveys: A case-control study in Guinea-Bissau. *Journal of Global Health*. 2023;13:04086. <https://doi.org/10.7189/jogh.13.04086>
22. Liu, Y., **Chu, Y.**, Yeung, D., Wang, W., Wang, L., Yin, P., Liu, J., Zhou, M., Liu, L. National and sub-national levels and causes of mortality among 5-19-year-olds in China in 2004-2019: A systematic analysis of evidence from the Disease Surveillance Points System. *Journal of Global Health*. 2022;12:11008. <https://doi.org/10.7189/jogh.12.11008> [joint first authorship]
21. Perin, J., **Chu, Y.**, Villavicencio, F., Schumacher, A., McCormick, T., Guillot, M., Liu, L. Adapting and validating the log quadratic model to derive under-five age- and cause-specific mortality (U5ACSM): a preliminary analysis. *Population Health Metrics*. 2022;20(1):3. <https://doi.org/10.1186/s12963-021-00277-w>
20. Turner, A. N., Kline, D., Norris, A., Phillips, W. G., Root, E., Wakefield, J., Li, Z., Lemeshow, S., Spahnie, M., Luff, A., **Chu, Y.**, et al. Prevalence of current and past COVID-19 in Ohio adults. *Annals of Epidemiology*. 2022;67:50-60. <https://doi.org/10.1016/j.annepidem.2021.11.009>
19. Schumacher, A. E., McCormick, T. H., Wakefield, J., **Chu, Y.**, Perin, J., Villavicencio, F., Simon, N., Liu, L. A flexible Bayesian framework to estimate age- and cause-specific child mortality over time from sample registration data. *Annals of Applied Statistics*. 2022;16(1):124-143. <https://doi.org/10.1214/21-AOAS1489>
18. Kline, D., Li, Z., **Chu, Y.**, Wakefield, J., Miller, W. C., Turner, A. N., Clark, S. J. Estimating seroprevalence of SARS-CoV-2 in Ohio: A Bayesian multilevel poststratification approach with multiple diagnostic tests. *Proceedings of the National Academy of Sciences of the United States of America*. 2021;118(26):e2023947118. <https://doi.org/10.1073/pnas.2023947118>
17. Chandramohan, D., Fottrell, E., Leita, J., Nichols, E., Clark, S. J., Alsokhn, C., Cobos Munoz, D., AbouZahr, C., Di Pasquale, A., Mswia, R., Choi, E., **Chu, Y.**, et al. Estimating causes of death where there is no medical certification: Evolution and state of the art of verbal autopsy. *Global Health Action*. 2021;14(sup1):1982486. <https://doi.org/10.1080/16549716.2021.1982486>
16. Helleringer, S., Liu, L., **Chu, Y.**, Rodrigues, A., Fisker, A. B. Biases in survey estimates of neonatal mortality: Results from a validation study in urban areas of Guinea-Bissau. *Demography*. 2020;57(5):1705-1726. <https://doi.org/10.1007/s13524-020-00911-6>
15. Wahl, B., Deloria Knoll, M., Shet, A., Gupta, M., Kumar, R., Liu, L., **Chu, Y.**, et al. National, regional, and state-level pneumonia and severe pneumonia morbidity in children in India:

- modelled estimates for 2000 and 2015. *Lancet Child and Adolescent Health*. 2020;4(9):678-687. [https://doi.org/10.1016/S2352-4642\(20\)30129-2](https://doi.org/10.1016/S2352-4642(20)30129-2)
14. **Chu, Y.** SVD-Bayes: A singular value decomposition-based approach under Bayesian framework for indirect estimation of age-specific fertility and mortality [master's thesis]. The Ohio State University; 2020. https://rave.ohiolink.edu/etdc/view?acc_num=osu1609638415015896
 13. McAllister, D. A., Liu, L., Shi, T., **Chu, Y.**, Reed, C., Burrows, J., Adeloye, D., et al. Global, regional, and national estimates of pneumonia morbidity and mortality in children younger than 5 years between 2000 and 2015: a systematic analysis. *Lancet Global Health*. 2019;7(1):e47-e57. [https://doi.org/10.1016/S2214-109X\(18\)30408-X](https://doi.org/10.1016/S2214-109X(18)30408-X)
 12. Liu, L., **Chu, Y.**, Oza, S., Hogan, D., Perin, J., Bassani, D. G., Ram, U., et al. National, regional, and state-level all-cause and cause-specific under-5 mortality in India in 2000-15: a systematic analysis with implications for the Sustainable Development Goals. *Lancet Global Health*. 2019;7(6):e721-e734. [https://doi.org/10.1016/S2214-109X\(19\)30080-4](https://doi.org/10.1016/S2214-109X(19)30080-4) [joint first authorship]
 11. Wahl, B., Sharan, A., Deloria Knoll, M., Kumar, R., Liu, L., **Chu, Y.**, McAllister, D. A., et al. National, regional, and state-level burden of *Streptococcus pneumoniae* and *Haemophilus influenzae* type b disease in children in India: modelled estimates for 2000-15. *Lancet Global Health*. 2019;7(6):e735-e747. [https://doi.org/10.1016/S2214-109X\(19\)30081-6](https://doi.org/10.1016/S2214-109X(19)30081-6)
 10. Choudhary, T. S., Sinha, B., Khera, A., Bhandari, N., **Chu, Y.**, Jackson, B., Walker, N., et al. Factors associated with the decline in under-five diarrhea mortality in India: a LiST analysis. *Journal of Global Health*. 2019;9(2):020804. <https://doi.org/10.7189/jogh.09.020804>
 9. Black, R., Fontaine, O., Lamberti, L., Bhan, M., Huicho, L., El Arifeen, S., Masanja, H., **Chu, Y.**, et al. Drivers of the reduction in childhood diarrhea mortality 1980-2015 and interventions to eliminate preventable diarrhea deaths by 2030. *Journal of Global Health*. 2019;9(2):020801. <https://doi.org/10.7189/jogh.09.020801>
 8. Countdown to 2030 Collaboration. Countdown to 2030: Tracking progress towards universal coverage for reproductive, maternal, newborn and child health. *Lancet*. 2018;391(10129):1538-1548. [https://doi.org/10.1016/S0140-6736\(18\)30104-1](https://doi.org/10.1016/S0140-6736(18)30104-1)
 7. Wahl, B., O'Brien, K. L., Greenbaum, A., Majumder, A., Liu, L., **Chu, Y.**, Luksic, I., et al. Burden of *Streptococcus pneumoniae* and *Haemophilus influenzae* type b disease in children in the era of conjugate vaccines: global, regional, and national estimates for 2000-15. *Lancet Global Health*. 2018;6(7):e744-e757. [https://doi.org/10.1016/S2214-109X\(18\)30247-X](https://doi.org/10.1016/S2214-109X(18)30247-X)
 6. He, C., Liu, L., **Chu, Y.**, Perin, J., Dai, L., Li, X., Miao, L., et al. National and subnational all-cause and cause-specific child mortality in China, 1996-2015: a systematic analysis with implications for the Sustainable Development Goals. *Lancet Global Health*. 2017;5(2):e186-e197. [https://doi.org/10.1016/S2214-109X\(16\)30334-5](https://doi.org/10.1016/S2214-109X(16)30334-5)
 5. Wilson, L. M., Rebholz, C. M., Jirru, E., Liu, M. C., Zhang, A., Gayleard, J., **Chu, Y.**, Robinson, K. A. Benefits and harms of osteoporosis medications in patients with chronic kidney disease: a systematic review and meta-analysis. *Annals of Internal Medicine*. 2017;166(9):649-658. <https://doi.org/10.7326/M16-2752>
 4. Liu, L., Oza, S., Hogan, D., **Chu, Y.**, Perin, J., Zhu, J., Lawn, J. E., et al. Global, regional, and national causes of under-5 mortality in 2000-15: an updated systematic analysis with

implications for the Sustainable Development Goals. *Lancet*. 2016;388(10063):3027-3035. [https://doi.org/10.1016/S0140-6736\(16\)31593-8](https://doi.org/10.1016/S0140-6736(16)31593-8)

3. Liu, L., Kalter, H. D., **Chu, Y.**, Kazmi, N., Koffi, A. K., Amouzou, A., Joos, O., Munos, M., Black, R. E. Understanding misclassification between neonatal deaths and stillbirths: empirical evidence from Malawi. *PLOS ONE*. 2016;11(12):e0168743. <https://doi.org/10.1371/journal.pone.0168743>
2. Song, P., Theodoratou, E., Li, X., Liu, L., **Chu, Y.**, Black, R. E., Campbell, H., Rudan, I., Chan, K. Y. Causes of death in children younger than five years in China in 2015: an updated analysis. *Journal of Global Health*. 2016;6(2):020802. <https://doi.org/10.7189/jogh.06.020802>
1. Maruthur, N. M., Tseng, E., Hutfless, S., Wilson, L. M., Suarez-Cuervo, C., Berger, Z., **Chu, Y.**, Iyoha, E., Segal, J. B., Bolen, S. Diabetes medications as monotherapy or metformin-based combination therapy for type 2 diabetes: a systematic review and meta-analysis. *Annals of Internal Medicine*. 2016;164(11):740-751. <https://doi.org/10.7326/M15-2650>

Working papers

2. **Chu, Y. et al.** Multimodal Artificial Intelligence for Cause-of-Death Assignment From Verbal Autopsy Records: Evaluation Across Two Independent Datasets
1. **Chu, Y. et al.** Sufficiency of Information in Population Surveys: Evidence from Verbal Autopsy

Technical reports and book chapters

3. Ministry of Public Health (Afghanistan), Royal Tropical Institute. Afghanistan Health Survey 2018. https://www.kit.nl/wp-content/uploads/2026/03/AHS-2018-report-final_30.11.2018_erratum12.02.2026.pdf
2. Liu, L., Hill, K., Oza, S., Hogan, D., **Chu, Y.**, Cousens, S., Mathers, C., Stanton, C., Lawn, J., Black, R. E. Levels and Causes of Mortality under Age Five Years. In: *Reproductive, Maternal, Newborn, and Child Health: Disease Control Priorities, Third Edition (Volume 2)*. Washington, DC: The World Bank; 2016. Chapter 4. https://doi.org/10.1596/978-1-4648-0348-2_ch4
1. Bolen, S., Tseng, E., Hutfless, S., Segal, J. B., Suarez-Cuervo, C., Berger, Z., Wilson, L. M., **Chu, Y.**, et al. *Medications for adults with type 2 diabetes: An update focused on monotherapy and add-on therapy to metformin* (EHC-2014-11-0075.R4). AHRQ Comparative Effectiveness Review Report No. 16-EHC013-EF. Rockville, MD: Agency for Healthcare Research and Quality; 2015. <https://www.ncbi.nlm.nih.gov/books/NBK362863/>

PRESENTATIONS

Posters are marked with [†].

Academic Presentations

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| May 2026 | <i>AI as Measurement Infrastructure: Adaptive Survey Design in Verbal Autopsy.</i> [†] FDS Workshop: AI for Social Science Research Methods. Yale University, New Haven, CT. |
| May 2026 | <i>Sufficiency of Information in Population Surveys: Evidence from Verbal Autopsy.</i> [†] Annual Meeting of the Population Association of America, St. Louis, MO. |
| May 2025 | <i>Leveraging language models and machine learning techniques for verbal autopsy analysis.</i> Gates Foundation, Dubai, United Arab Emirates. Invited talk. |
| Apr 2025 | <i>Automating cause of death classification from verbal autopsy: Using pre-trained language models and multimodal learning.</i> Annual Meeting of the Population Association of America, Washington, DC. |
| Nov 2024 | <i>Leveraging machine learning techniques for verbal autopsy analysis.</i> Demystifying Machine Learning Workshop, Max Planck Institute for Demographic Research, Rostock, Germany. |
| Apr 2023 | <i>Temporal changes in cause of death among adults in six countries in Eastern and Southern Africa: A multi-country cohort study using verbal autopsy data.</i> Annual Meeting of the Population Association of America, New Orleans, LA. |
| May 2021 | <i>Indirect estimation of age-specific fertility and mortality: SVD-Bayes model.</i> Annual Meeting of the Population Association of America, Virtual. |
| Nov 2018 | <i>Global estimates for meningitis-specific mortality.</i> Meningitis Disease Burden Evaluation Alignment Meeting of the World Health Organization and Bill and Melinda Gates Foundation, Seattle, WA. Invited talk. |
| Apr 2017 | <i>Understanding misclassification between neonatal deaths and stillbirths: Empirical evidence from Malawi.</i> Annual Meeting of the Population Association of America, Chicago, IL. |
| Apr 2017 | <i>National and subnational levels and causes of child mortality in China, 1996-2015: Implications for the SDGs.</i> Annual Meeting of the Population Association of America, Chicago, IL. |
| Sep 2016 | <i>Estimating national and subnational causes of child deaths using data from the China Maternal and Child Health Surveillance System.</i> Sixth Child Survival Forum in China, Chengdu, China. |
| Aug 2016 | <i>Understanding misclassification between neonatal deaths and stillbirths: Empirical evidence from Malawi.</i> WHO Informal Technical Consultation on Maternal and Neonatal Deaths and Stillbirths, Geneva, Switzerland. |

- Apr 2016 *Global, regional, and national causes of child mortality in 2000-2015: Reflecting on MDG 4 and prioritizing for SDG 3.2.* Annual Meeting of the Population Association of America, Washington, DC.
- Apr 2015 *Progress in estimating causes of death among children aged 1-59 months – Update for 2000-2015.* Maternal and Child Epidemiology Estimation Project Review Meeting, Geneva, Switzerland.
- Apr 2015 *Differences in cause-of-death distribution estimates in China: MCMSS vs MCEE.* Maternal and Child Epidemiology Estimation Project Review Meeting, Geneva, Switzerland.

Contributed presentations

- May 2021 *Changes in age at last birth over the course of contemporary fertility declines.* Annual Meeting of the Population Association of America, St. Louis, MO.
- Apr 2019 *Understanding reporting errors of population survey-based neonatal mortality: A validation study from Guinea-Bissau.* Annual Meeting of the Population Association of America, Austin, TX.
- Apr 2018 *Cause-specific mortality rates for fine age groups among children under five.* Annual Meeting of the Population Association of America, Denver, CO.
- Dec 2017 *China Maternal and Child Health Surveillance System.* Global Age Patterns of Under-Five Mortality Project Kickoff Meeting, Philadelphia, PA.
- Aug 2017 *Age-specific child causes of death.* IUSSP International Seminar on Mortality: Past, Present and Future, Campinas, Brazil.
- Aug 2016 *Estimating national and subnational causes of child deaths using data from the China Maternal and Child Health Surveillance System.* WHO Informal Technical Consultation on Maternal and Neonatal Deaths and Stillbirths, Geneva, Switzerland.
- Jun 2016 *Global burden of Streptococcus pneumoniae in children under 5 in the era of PCV: 2000-2015.* International Symposium on Pneumococci and Pneumococcal Disease, Glasgow, Scotland.
- Jun 2016 *Mortality caused by Streptococcus pneumoniae in children under 5 in India: A state-by-state analysis from 2000 to 2015.* International Symposium on Pneumococci and Pneumococcal Disease, Glasgow, Scotland.
- Dec 2015 *Subnational child cause of death estimation in China.* UN IGME Technical Advisory Group Bi-annual Meeting, Washington, DC.
- Nov 2015 *Causes of child mortality in 2000-2015: The global picture and current state in Nigeria.* Nigeria Verbal/Social Autopsy Technical Working Group Meeting, Abuja, Nigeria.
- May 2015 *Progress in estimating causes of child mortality in 2000-2015.* UN IGME Technical Advisory Group Bi-annual Meeting, New York, NY.

Department Seminars

Feb 2026	<i>Leveraging language models and machine learning techniques for verbal autopsy analysis.</i> Computational Demography Working Group Seminar, University of Washington, Seattle, WA.
Apr 2025	<i>Automating cause of death classification.</i> HEALMOD Professional Seminar, The Ohio State University, Columbus, OH.
Apr 2021	<i>Indirect estimation of age-specific fertility and mortality: SVD-Bayes model.</i> [†] Hayes Graduate Research Forum, Columbus, OH.
Oct 2021	<i>Indirect estimation of age-specific fertility and mortality: SVD-Bayes model.</i> TDAI Fall Forum, The Ohio State University, Columbus, OH.

GRANT PARTICIPATION

Aug 2022–Jan 2025	Reference Data Archive (no-cost extension to Jan 2026) Funding: The Ohio State University (Prime: Bill and Melinda Gates Foundation) Principal Investigator: Samuel Clark. Role: Key personnel
Apr 2022–May 2023	Development and Validation of Verbal Autopsy Algorithms Using openVA Tools Funding: The Ohio State University (Prime: CDC Foundation) Principal Investigator: Samuel Clark. Role: Graduate Research Assistant
May 2021–Dec 2021	COVID-19 Rapid Mortality Surveillance Methods Development Funding: The Ohio State University (Prime: Vital Strategies) Principal Investigator: Samuel Clark. Role: Graduate Research Assistant
Dec 2009–Apr 2011	Student Principal Investigator , Department of Maternal and Child Health, Fudan University, Shanghai, China

CONSULTANCY

May 2023–present	Consultant, Reference Data Archive project , World Health Organization
Dec 2022–Jan 2023	Consultant , Setting up survey ODK data collection and centralized data system. Ministry of Health, Guinea-Bissau
Apr 2021–May 2021	Consultant , WHO VARG Workshop / verbal autopsy instrument revision, World Health Organization
Jul 2018–Aug 2018	Consultant , Review data quality in Global Health Estimates portal. Maternal, Newborn, Child and Adolescent Health Division, World Health Organization, Geneva, Switzerland
Jul 2018–Aug 2018	Consultant , Mortality and cause-specific mortality modeling for Afghanistan Health Survey. KIT Royal Tropical Institute, Amsterdam, Netherlands

TEACHING AND MENTORING

Teaching

SP2026	SOCIOL 8607 (OSU) , Teaching Assistant for Causal Modeling
AU2021	SOCIOL 3549 (OSU) , Recitation Leader for Statistics in Sociology
AU2016–AU2017	380.651 (JHSPH) , Teaching Assistant for Methods and Measures in Population Studies
AU2013–AU2014	380.603 (JHSPH) , Teaching Assistant for Demographic Methods in Public Health

Research Supervision and Mentoring

AU2025	Provided day-to-day technical supervision to a PhD research assistant on data quality assessment, data cleaning, and coding for the Reference Data Archive project, The Ohio State University
SP2024	Mentored an undergraduate research assistant, The Ohio State University
SU2023	Provided day-to-day technical supervision to a PhD research assistant on data quality assessment, data cleaning, and coding for the Reference Data Archive project, Africa Health Research Institute
2015–2018	Provided day-to-day research supervision and technical mentoring to 10+ graduate research assistants across literature reviews, data extraction, data analysis, modeling support, and data visualization, JHSPH
SU2017, SU2018	Mentored two undergraduate students, JHSPH Diversity Summer Internship Program

Workshops

2022	Workshops on R hosted by Sociology Graduate Student Association
2018–2020	Verbal autopsy data analysis workshops

PROFESSIONAL ACTIVITIES

Society Membership

2015–present	Population Association of America (PAA)
2019–present	International Union for the Scientific Study of Population (IUSSP)

Center Affiliations

2018–present	Institute for Population Research, OSU
2018–present	Translational Data Analytics Institute, OSU
2024–2026	Health and Environment Modeling Laboratory, OSU
2022–2023	Nationwide Center for Advanced Customer Insights, OSU
2015–2018	Johns Hopkins Institute for International Programs, JHSPH
2014	Johns Hopkins Evidence-based Practice Center, JHSPH
2013–2014	Johns Hopkins Center for Adolescent Health, JHSPH

Conference Organization

2025	Session Chair and Discussant, Annual Meeting of the Population Association of America
2021	Session Chair, Annual Meeting of the Population Association of America
2021	Session Chair, Annual Meeting of Applied Linguistics and Professional Practice
2016	The Sixth Child Survival Forum in China, Chengdu, China
2013	Technical Advisory Group Annual Meeting on High Impact Practices in Family Planning, UNFPA, New York, USA

EDITORIAL ACTIVITIES

Peer Review

PNAS; Demographic Research; Population Health Metrics; Annals of Epidemiology; Nature; Journal of Global Health; Preventing Chronic Disease; European Journal of Population; PLOS ONE; PLOS Global Health; Clinical Epidemiology; BMC Public Health; BMC Pediatrics; Journal of Epidemiology and Global Health; Frontiers; Population; Clinical Epidemiology and Global Health.

HONORS AND AWARDS

2018, 2023	Distinguished University Fellowship, The Ohio State University
2013	Tuition Scholarship, Johns Hopkins Bloomberg School of Public Health
2011–2012	People's Scholarship of Fudan University
2007–2009	People's Scholarship of Fudan University

PROFESSIONAL DEVELOPMENT

2025	PAA workshop: A Primer on Machine Learning and Artificial Intelligence for Population Research
2024	Max Planck Institute for Demographic Research workshop: Demystifying Machine Learning
2022	Berkeley Population Center Annual Workshop on Formal Demography
2021	Pedagogical training with The Drake Institute for Teaching and Learning, Columbus, USA
2019	Data Carpentry workshop in Social Sciences, Columbus, USA
2018	D4H/WHO workshop on Verbal Autopsy Methods, Columbus, USA
2018	IUSSP workshop: Bayesian Small Area Estimation using Complex Survey Data: Methods and Applications, Denver, USA
2018	Survey Research Design, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
2017	PAA workshop: Social Media and Demographic Methods, Chicago, USA
2017	Methods and Measures in Population Studies, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
2016	PAA workshop: Bayesian Population Projections: Theory and Practice, Washington, DC, USA
2015	Analysis of Longitudinal Data, Johns Hopkins Bloomberg School of Public Health, Baltimore, USA

SKILLS

Statistical software	R, Python, Julia, Stata, SPSS, SQLite/SQL, RJAGS, RNIMBLE, JMP
Data science toolkits	Hugging Face, Optuna, prompt engineering, Azure, Kubeflow, Snowflake, Teradata, DBeaver, Amazon Web Services (AWS/S3)